

5G/6G Communication Systems- ECA5601 (LAB)

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Experiment 13: V2X Latency Comparison

Aim: To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Objective 1:

Program:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])

figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
```

% ----- Graph 2 -----

subplot(2,1,2)

plot(1:length(latency),latency,'-s','LineWidth',2)

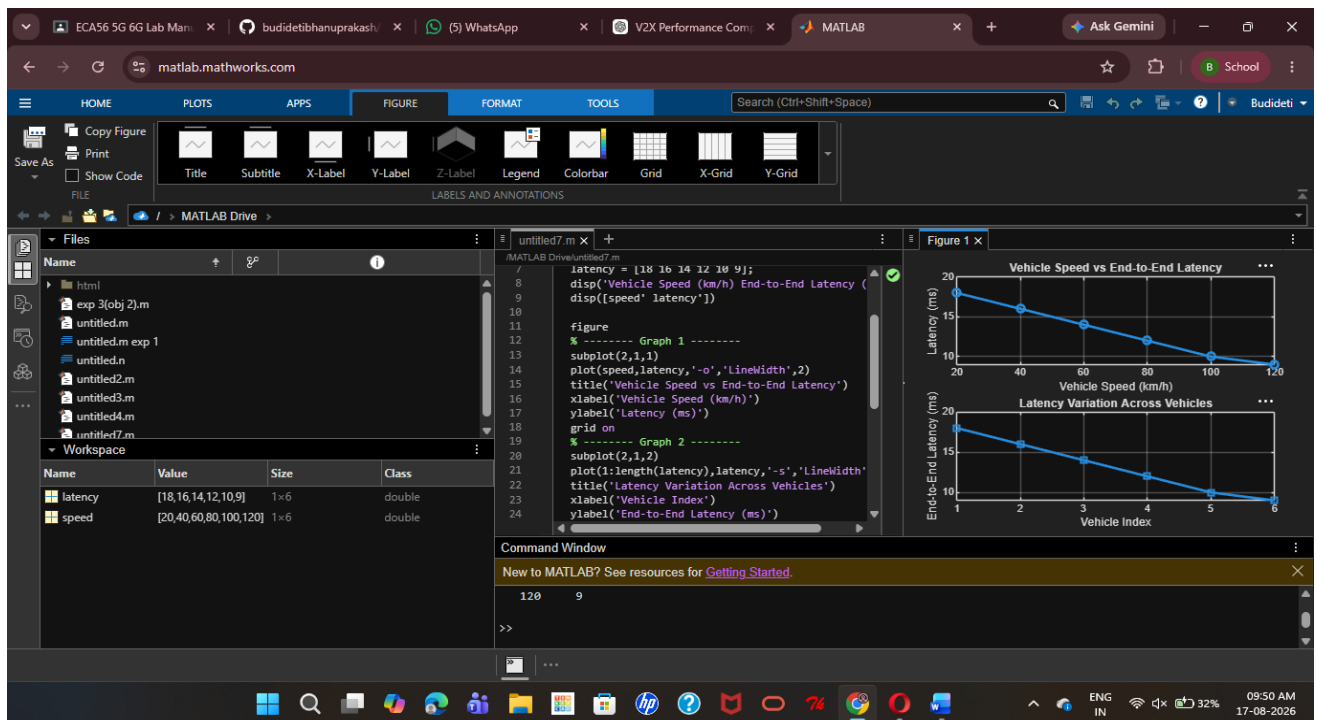
title('Latency Variation Across Vehicles')

xlabel('Vehicle Index')

ylabel('End-to-End Latency (ms)')

grid on

Output:



Objective 2:

Program:

clc;clear;close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')

disp([speed' PDR'])

figure

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(speed,PDR,'-o','LineWidth',2)
```

```
title('Vehicle Speed vs Packet Delivery Ratio')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(1:length(PDR),PDR,'-s','LineWidth',2)
```

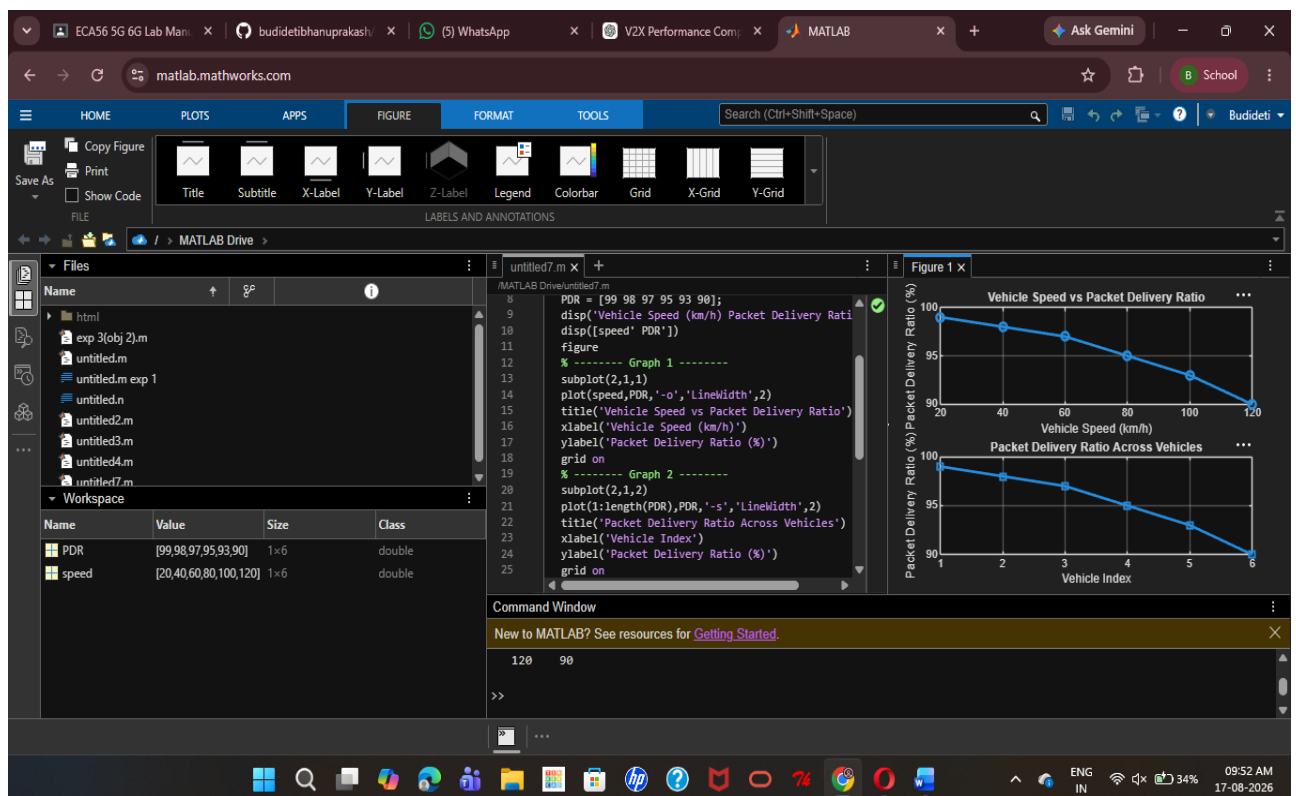
```
title('Packet Delivery Ratio Across Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

Output:

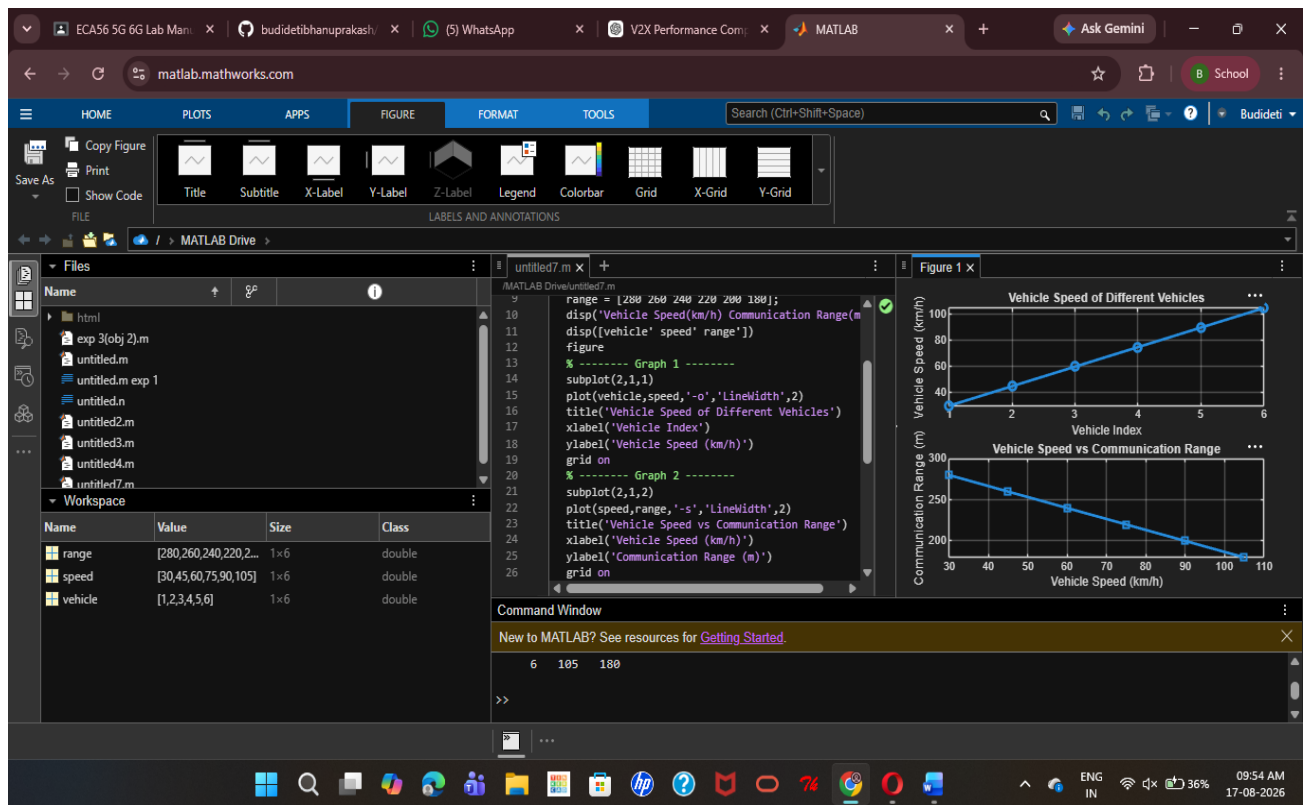


Objective 3:

MATLAB Code:

```
clc;
clear;
close all;
% Vehicle IDs
vehicle = 1:6;
% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on
```

Output:



Result:

The V2X system was successfully simulated, and the effects of latency, packet delivery ratio, and vehicle speed on communication performance were analyzed.